

KOTLAS, Russian Federation

WMO: 228870

Lat: 61.2395N

Lon: 46.6950E

Elev: 183

StdP: 14.60

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.0	-20.9	-31.1	1.0	-26.8	-25.0	1.4	-20.9	18.7	20.7	17.4	21.6	2.8	330	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	83.6	69.0	79.8	67.1	76.5	65.1	71.0	80.5	68.8	77.4	66.7	74.0	7.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.6	102.2	76.1	65.6	95.0	73.6	63.6	88.4	71.3	34.9	80.9	33.1	77.5	31.4	73.6	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.2	15.2	13.5	-32.3	88.4	6.3	3.9	-36.8	91.3	-40.5	93.6	-44.0	95.8	-48.6	98.7		
			-32.1	74.2	6.5	2.4	-36.7	75.9	-40.5	77.3	-44.1	78.7	-48.8	80.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.2	9.3	11.3	22.9	36.8	49.2	57.7	63.3	58.0	48.7	37.0	23.6	14.9
	DBStd	21.38	15.25	15.26	10.31	8.83	9.37	8.06	6.98	7.33	7.03	7.67	11.93	14.78
	HDD50	6147	1261	1083	839	404	135	20	1	12	106	408	791	1087
	HDD65	10626	1726	1503	1304	845	496	244	118	240	489	868	1241	1552
	CDD50	1115	0	0	0	9	109	252	412	261	67	5	0	0
	CDD65	122	0	0	0	0	6	27	65	23	1	0	0	0
	CDH74	1272	0	0	0	4	98	307	662	197	4	0	0	0
	CDH80	298	0	0	0	0	17	72	157	52	0	0	0	0
Wind	WSAvg	6.6	6.6	6.6	7.1	6.9	7.3	6.4	5.6	5.6	6.2	6.9	6.7	6.9
Precipitation	PrecAvg	21.2	1.3	0.9	0.9	1.4	1.7	2.4	2.8	2.6	2.0	2.0	1.6	1.5
	PrecMax	29.4	2.2	2.1	2.0	4.3	3.7	5.9	5.7	5.9	3.7	5.1	3.6	3.0
	PrecMin	15.8	0.3	0.1	0.1	0.1	0.6	0.6	0.7	0.7	0.5	0.2	0.5	0.4
	PrecStd	3.0	0.5	0.4	0.5	0.8	0.7	1.2	1.2	1.3	0.9	0.8	0.6	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.9	36.9	47.2	71.3	82.5	87.6	88.7	86.0	73.4	61.0	47.0	39.4
		MCWB	33.7	34.8	38.7	56.1	65.6	71.3	72.6	71.1	63.3	52.8	45.4	38.4
	2%	DB	33.4	34.6	41.6	61.3	76.5	81.4	84.3	78.9	67.2	54.3	41.1	35.8
		MCWB	32.4	33.0	35.9	49.8	61.6	67.1	70.0	67.4	59.2	50.1	40.0	34.7
	5%	DB	31.8	33.1	38.3	55.2	71.6	77.2	80.8	74.7	63.3	50.6	38.6	34.1
		MCWB	30.9	31.9	34.3	45.7	58.2	64.9	68.6	65.3	56.6	47.6	37.6	33.1
	10%	DB	28.7	30.7	36.0	50.1	66.3	73.2	77.4	70.5	59.6	47.6	36.0	32.8
		MCWB	27.6	29.4	32.9	42.3	55.2	62.3	66.4	63.0	55.0	45.5	35.0	31.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.1	35.6	39.8	56.2	67.5	72.5	74.1	72.7	64.1	54.2	45.4	38.4
		MCDB	34.4	36.5	44.3	71.1	77.7	83.5	84.6	82.9	71.8	57.8	46.6	39.1
	2%	WB	32.7	33.5	37.2	50.3	63.0	69.3	71.6	69.4	60.6	50.4	40.1	35.0
		MCDB	33.4	34.3	40.4	60.3	73.6	78.7	81.4	77.5	65.5	53.5	40.8	35.5
	5%	WB	30.8	31.9	35.0	46.3	59.6	66.6	69.6	66.6	57.9	47.8	37.6	33.3
		MCDB	31.7	32.9	37.6	54.0	68.7	74.3	78.5	72.6	61.9	50.2	38.3	33.9
	10%	WB	27.6	29.5	33.3	43.3	56.2	63.9	67.6	63.9	55.5	45.5	35.2	31.9
		MCDB	28.6	30.7	35.5	49.1	65.2	71.6	75.9	69.1	59.3	47.6	35.9	32.8
Mean Daily Temperature Range	5% DB	MDBR	11.7	13.3	15.6	16.8	19.7	19.0	19.3	17.0	14.2	8.6	7.9	9.8
		MCDBR	10.2	9.2	15.6	25.3	28.2	25.7	25.0	22.8	20.3	14.3	8.0	7.0
		MCWBR	10.3	8.6	11.1	15.7	15.2	13.0	12.5	12.1	12.6	10.3	7.6	6.8
	5% WB	MCDBR	10.0	8.7	11.9	23.2	25.0	22.7	22.9	20.3	17.6	13.1	7.5	6.7
		MCWBR	10.1	8.5	9.3	15.0	14.5	12.5	12.3	11.7	11.6	10.2	7.3	6.8
Clear-Sky Solar Irradiance	taub	0.255	0.300	0.334	0.366	0.358	0.357	0.391	0.375	0.343	0.322	0.275	0.227	
	taud	2.011	2.058	2.064	2.195	2.432	2.462	2.370	2.395	2.468	2.362	2.109	1.950	
	Ebn at Noon	138	205	241	260	275	277	262	257	246	205	131	92	
	Edh at Noon	13	24	35	37	32	31	34	30	24	18	12	8	
All-Sky Solar Radiation	RadAvg	92	305	708	1177	1566	1661	1688	1171	656	258	80	43	
	RadStd	13	69	113	140	110	172	148	128	79	33	16	8	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+1.04	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+1.06	N/A	N/A	N/A	N/A	+1.15	N/A	N/A	N/A	N/A

Nomenclature: See separate page